

1. Project Setup

Goal: Initialize the backend application where all logic is handled on the server

Tasks:

- Create repository: `tic-tac-toe`
- Configure backend environment:
 - Set up Python with a virtual environment
 - Install dependencies: `fastapi`, `uvicorn`, `pydantic`

2. Data Storage

Goal: Store and manage all application data on the server using JSON

Tasks:

- Implement server-side persistence methods:
 - `load_state()` – load JSON data when the server starts
 - `save_state()` – persist data after every state change
- Handle errors on the server:
 - If file is missing → initialize empty state
- Define server-managed data structure:
 - `players`
 - `matches`
 - `statistics`

3. Application Management (Backend Logic)

Goal: Centralize all business logic on the server

Tasks:

- Implement `state.py` (server-side state manager):
 - Maintain in-memory state on the server
 - Synchronize state with JSON storage
- Provide backend operations:
 - Create player
 - Delete player

- Create match
- Update match

4. Game Logic

Goal: Execute all gameplay rules on the server

Tasks:

- Represent the board as an array of 9 fields (server-side)
- Implement server-controlled logic:
 - Player moves
 - Turn switching
 - Move validation
- Detect game results on the server:
 - Use predefined WIN_PATTERNS
- Handle outcomes:
 - Win
 - Draw
 - Loss
- Update player statistics on the server

5. API (Client - Server Communication)

Goal: Expose server-side functionality via HTTP endpoints

Endpoints:

- POST /startMatch
- POST /updateMatch
- POST /createPlayer
- DELETE /deletePlayer
- GET /state

Tasks:

- Implement routes in `routes.py`
- Validate all incoming requests using Pydantic (server-side validation)
- Ensure all operations are processed on the server, not the client

6. Dashboard (Server Data Visualization)

Goal: Provide visibility into server-managed data

Tasks:

- Build a CLI tool that interacts with server data:
 - Read JSON storage
 - Display:
 - Players
 - Statistics
 - Active matches

7. Validation

Goal: Ensure server-side data persistence across restarts

Tasks:

- Restart the backend server
- Verify that stored data is preserved:
 - Players
 - Matches
 - Statistics